

PawLink — Tech for Good Vision Report

AI-Powered Animal Rescue Intelligence Platform

Prepared for HackDelft 2026 · IMC IM CHANGE Challenge · Two-Person CESE Team

Executive Summary

Every month, a single branch of the Dierenambulance & Hospitaal Den Haag network handles close to 1,000 injured or found animals — many of them unidentified and untracked. In 2025 alone, the Den Haag service answered over 44,000 calls and dispatched ambulances more than 18,000 times. Roughly 1,400 animals were sheltered in their own hospital that year, with 220 recovered animals successfully re-homed. Behind each of these numbers is a cascade of manual, fragmented processes: phone-based intake, manual chip scanning, cross-referencing multiple databases, paper-based care logs, and manual outreach to find owners or adopters.^{[1][2][^3]}

PawLink is a modular, open-source intelligence layer for animal rescue organizations — beginning with an AI-first intake, identification, and owner-reunification engine, but designed to scale into a full operational platform for the sector. It directly addresses the IMC IM CHANGE "Tech for Good" challenge, targeting registered non-profit animal welfare organizations (ANBI-status eligible in the Netherlands).

The name reflects its core mission: **forging the missing link between a found animal and its owner, its care, and its future home.**

The Problem Space

1. The Scale of Unidentified Animals

The Netherlands is home to approximately 1.7 million dogs and 3 million cats. While mandatory microchipping for dogs was introduced in 2013, cats have only just come under a mandatory chipping law effective 2026. This means that for years — and still today during the transition — a substantial share of incoming cats and stray animals arrive at rescue organizations **without any form of machine-readable identification.**^{[4][5][6][7][^8]}

The current identification workflow, when a chip is absent, relies on:

- Manual breed/color descriptions typed into Amivedi (the national lost-and-found registry) ^[^9]
- Cross-checking the NDG (Nederlandse Databank Gezelschapsdieren) and PetBase separately^{[10][11]}
- Phone calls to neighboring shelters

- Passive waiting for owners to self-report

Even when a chip exists, multiple databases (NDG, PetBase, Amivedi, Raad van Beheer) do not integrate seamlessly. The Dutch chip checker portal `chipnummer.nl` aggregates lookups but cannot proactively push notifications to owners.^{[12][11]}

2. The Fragmented Ecosystem

The lost pet journey in the Netherlands involves: the animal ambulance, the shelter, Amivedi, the chip database, the municipality, the vet, and social media platforms (Facebook groups, WhatsApp neighborhood apps, [PetAlert.nl](#)). Each operates in isolation. An owner must manually contact all of them. The burden of coordination falls entirely on the stressed owner — or worse, on no one at all.^{[13][14][15]}

Nationally, Amsterdam-based CatRadar achieves a 68% success rate using geo-targeted Facebook and Instagram ads — essentially manual digital marketing. This shows that better targeted outreach works, but the approach doesn't scale or integrate with operational shelter workflow.^[16]

3. The Operational Gap Inside Shelters

Current shelter management software (e.g., Salesforce-based solutions, spreadsheets) is generic and not purpose-built for the triage + care + re-homing pipeline of a receiving animal ambulance. Specifically:^{[17][18]}

- **No AI-assisted triage at intake:** Injury assessment is entirely manual.
- **No predictive capacity planning:** Intake volumes fluctuate seasonally but shelters lack forecasting tools.^{[19][20]}
- **No automated adoption matching:** Matching is based on manual conversations, not data-driven profiling.^[19]
- **No ID campaign coordination:** Microchipping campaigns are run ad-hoc, without network-level coordination or impact tracking.^[19]

4. Biometric Identification: The Gap Between Research and Practice

Academic and commercial research shows strong viability of biometric pet identification:

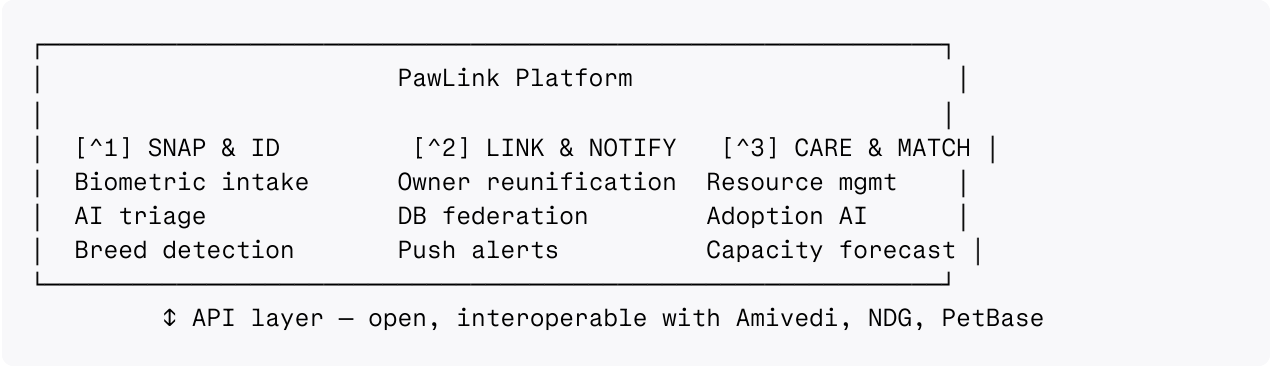
- Dog noseprint recognition achieves **98% accuracy** with deep neural networks using only a smartphone camera^{[21][22]}
- Pet facial recognition (Finding Rover) claims **98% accuracy**^{[23][22]}
- Ear vessel pattern biometrics have been studied for rodents^[24]
- Multi-modal frameworks combining facial features, coat pattern, ear shape, and body markings are now feasible^{[25][26]}

Despite this, **no rescue organization in the Netherlands currently deploys AI biometrics at intake**. The gap between research-grade tools and practical rescue workflows is entirely unbridged.

The Vision: PawLink

PawLink is an open AI platform that gives every found animal a digital identity within minutes of intake — using multimodal vision, database federation, and smart notifications — so no animal stays lost and no shelter stays overwhelmed.

Core Modules (The System Architecture)



Module 1: SNAP & ID — Multimodal Animal Identification at Intake

What It Does

A rescue worker photographs a newly arrived animal with a smartphone. Within seconds, PawLink runs:

1. **Breed & species classification** (transfer learning on VGG/ResNet, ~133 dog breeds available in open datasets)^{[27][28]}
2. **Facial feature embedding** (similar to Finding Rover's approach)^{[29][23]}
3. **Noseprint biometric** — prompted photo guidance for dogs, achieving near-98% identification accuracy in literature^{[30][21]}
4. **Coat pattern & markings extraction** (unique identifier for cats and mixed-breed dogs)
5. **Visual injury triage score** — a basic severity rating (limping, wound visible, labored breathing) informed by computer vision, similar to emerging tools like [VetVisit.io](#) and Pearl AI^{[31][32]}

The result is a **PawLink Digital Passport**: a structured profile combining all visual biometrics, automatically cross-matched against the lost-pet database populated by owners.

Why This Is Novel

Existing apps (Finding Rover, PetBiometrics) are consumer-facing: owners register lost pets and shelters optionally connect. **PawLink flips this**: it is shelter-first and intake-initiated. Every found animal gets a biometric record *proactively*, even before any owner reports it missing. This changes the paradigm from reactive search to proactive identification.^{[21][29]}

CESE Technical Stack

- **CS:** Python/FastAPI backend, fine-tuned MobileNet V3 or EfficientNet-Lite for on-device speed, OpenAI Vision API for triage description (the hackathon provides OpenAI access)
- **EC:** Edge-optimized inference path for low-connectivity shelter environments; RFID/NFC chip scanner integration (134.2 kHz ISO standard) as a companion hardware module in the longer term^[43]

Module 2: LINK & NOTIFY — Federated Database & Smart Owner Alerts

The Database Fragmentation Problem

In the Netherlands, a found animal's chip number must be cross-checked across NDG, PetBase, chipnummer.nl, and Amivedi — four separate lookups, none of which auto-notify owners. PawLink creates a **federated query layer**: one API call checks all registered databases and initiates an immediate notification cascade if a match is found.^{[34][11][10]}

For Unchipped Animals (The Core Gap)

When no chip exists, PawLink queries its own **visual identity database** — populated by owners who pre-register their pets via a companion mobile app (or web form). Similarity matching against the biometric passport triggers automatic alerts to owners in the geographic proximity of the intake location.

Notification Cascade

Upon a match or high-confidence visual similarity:

1. **Push notification** to registered owner (mobile app / SMS / email)
2. **Geo-targeted social alert** pushed to PetAlert.nl-style community channels^[15]
3. **Automated WhatsApp/Telegram bot message** to local neighborhood groups (via API)
4. **Amivedi auto-report creation** via API if not already registered

This mirrors the multi-channel approach proven effective by CatRadar (68% success) but fully automates it and integrates it into shelter workflow.^[16]

Regulatory Alignment

The Dutch mandatory cat microchipping law (2026) and EU pet passport requirements create a **data compliance tailwind**: PawLink's architecture is designed to be GDPR-compliant and interoperable with RVO-approved databases.^{[6][7][4][10]}

Module 3: CARE & MATCH — Shelter Resource Intelligence

Predictive Intake Capacity

Time-series forecasting (ARIMA or LSTM on historical intake data) predicts weekly animal volumes by species and injury type, enabling proactive staffing and supply allocation. The Den Haag service's own data (18,000+ ambulance responses in 2025, 1,400 annual hospital intakes) provides a real training baseline.^{[20][2][^3]}

Personalized Care Tracking

Each PawLink Passport auto-generates a care timeline: injury status, medical procedures, feeding, behavior notes. This replaces paper logs with a structured, queryable system — enabling pattern detection (e.g., which injuries correlate with longer recovery, which breeds have higher re-homing rates).

AI Adoption Matching

When an animal is healthy and ready for adoption, PawLink generates a behavioral + medical profile and matches it against registered prospective adopters, analyzing lifestyle compatibility — similar to commercial adoption AI platforms but purpose-built for rescue organizations and free at the point of use.^[^19]

Microchipping Campaign Coordination

PawLink tracks **chipping** rates by neighborhood, species, and intake source, identifying "cold spots" where unchipped animals cluster. This powers data-driven ID campaign targeting — a direct amplifier for the Dutch government's 2026 cat microchipping initiative.^{[7][19]}

Current State of the Art & Gaps Addressed

Area	Current Best Practice	Gap	PawLink's Contribution
Pet ID (chipped)	Manual chip scan + multi-DB lookup	No auto-notify, no single federation	Federated API + instant owner push
Pet ID (unchipped)	Visual description in Amivedi ^[^34]	Text-only, no matching, no proactive alert	AI visual biometrics + similarity search
Intake triage	Manual veterinary assessment	No decision support for volunteers	Vision-based injury severity pre-triage ^{[31][32]}
Owner reunification	Owner manually contacts 5+ organizations ^[^14]	Fragmented, owner-burden model	Automated multi-channel alert cascade
Shelter capacity	Ad-hoc, experiential	No forecasting, reactive resourcing	Time-series prediction on intake history ^[^20]
Adoption matching	Manual interview-based	No data-driven compatibility scoring	AI profile matching for adopter-animal fit ^[^19]

Area	Current Best Practice	Gap	PawLink's Contribution
ID campaign targeting	Generic, blanket outreach	No geographic heat-mapping of gaps	Intake analytics → targeted chipping campaigns

Why This Qualifies for IM CHANGE

The IMC IM CHANGE challenge requires using technology to create positive social impact for eligible non-profit organizations . PawLink maps directly to this:

- **Eligible organization type:** Dierenambulance & Hospitaal Den Haag is a foundation (Stichting) providing public benefit animal welfare services — aligned with ANBI-status requirements^[35]^[36]
- **Apolitical, non-religious:** Animal welfare is a clearly non-partisan humanitarian cause
- **Judging criteria alignment** (Technical Solution 1–10, Impact: Reach + Depth 1–10, Presentation 1–10) :
 - *Technical:* Multimodal AI, federated APIs, real-time notifications — strong CS/EC depth
 - *Impact Reach:* Chain-level deployment → potentially all ~3,000+ veterinary businesses in the Netherlands; European scalability^[^37]
 - *Impact Depth:* Directly reduces animal suffering time and increases owner reunification rates

Impact Assessment

Immediate (One Rescue Organization)

- A single Den Haag branch handling ~1,000 intakes/month → estimated 30–40% reduction in time-to-owner-reunification for unchipped animals (based on biometric match accuracy of ~95–98%)^[22]^[21]
- Volunteer hours saved on manual DB lookup: estimated 15–20 min/intake × 1,000/month = 250–330 staff-hours/month

Medium Scale (Netherlands Network)

- ~3 million cats now subject to mandatory chipping → PawLink's visual ID serves as a bridge during the transition years where compliance is still building^[8]^[7]
- National deployment across Amivedi's network would make it a de-facto national platform^[^9]

Long-Term / "Galen" Vision

PawLink's architecture is explicitly designed for extensibility:

- **Healthcare integration:** The injury triage module is a stepping stone toward AI-assisted veterinary decision support — connecting to telemedicine platforms and vet networks^[38]^[39]
- **Insurance:** Biometric identity is the prerequisite for per-animal health insurance enrollment (as used by Ping An in China)^[^22]
- **EU-wide interoperability:** With EU pet passport harmonization underway, PawLink's open API can serve cross-border reunification^[40]^[4]
- **Wildlife extension:** The same computer vision stack applies to wildlife rescue (birds, hedgehogs, wildlife — 5,000+ wild birds transported by Den Haag alone in 2024)^[^41]

Hackathon 24-Hour Execution Plan

Pre-Hackathon Preparation (Do This Now)

- [] Pre-train or fine-tune a breed classifier on the Stanford Dogs dataset + Oxford Pets (available open-source)
- [] Stand up a simple Supabase or Firebase backend for the pet passport schema
- [] Prepare a fake-but-realistic intake demo dataset (10–15 animal records)
- [] Design the demo narrative: one rescue worker, one found dog, zero chip — full flow in 90 seconds

Hour-by-Hour Build Plan (24 hours)

Phase	Hours	Deliverable
Setup & scaffold	0–2	FastAPI backend + React/Next.js frontend skeleton
Module 1: Vision ID	2–8	Photo upload → breed + visual hash → display passport
Module 2: DB federation	8–13	Simulated Amivedi/NDG query + owner notification (email/SMS via Twilio)
Module 3: Triage score	13–17	GPT-4o Vision triage description + severity badge on passport
Dashboard & UX polish	17–21	Shelter intake dashboard, search, match confidence display
Demo prep & pitch	21–24	Slides, demo video clip, impact numbers

Technology Choices for Speed

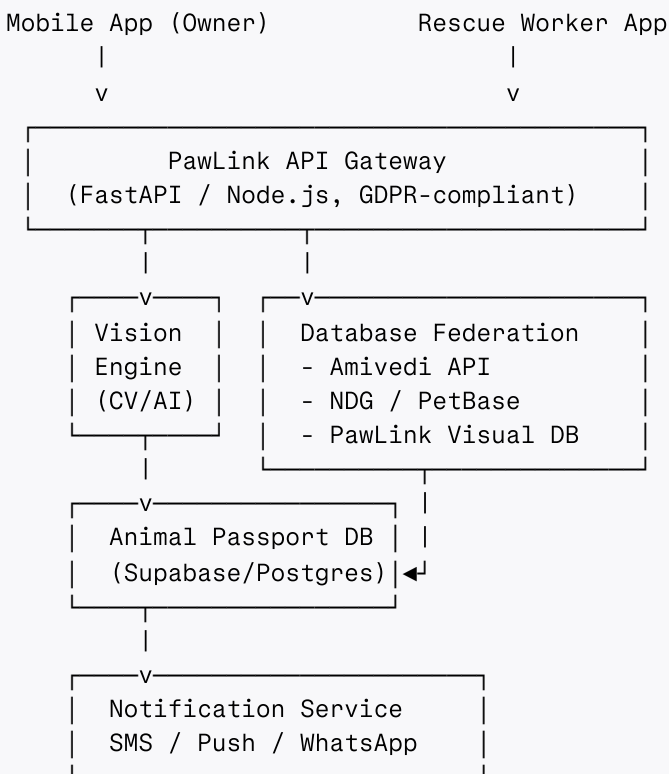
- **Backend:** Python + FastAPI (familiar to CS engineers)
- **AI:** OpenAI GPT-4o Vision API (provided by hackathon) for triage + description; HuggingFace pre-trained ViT for breed classification
- **Frontend:** React + Tailwind CSS or Lovable.dev for rapid UI scaffolding

- **Database:** Supabase (Postgres + real-time + storage in one)
- **Notifications:** Twilio SMS / SendGrid email (both have free tiers)
- **Hosting:** Vercel (frontend) + Railway/Render (backend) — deploys in minutes

Demo Script for Judges

1. (0:00) Show a photo of a found dog arriving at a shelter (no chip)
2. (0:20) Drag photo into PawLink intake screen → passport generates in 3 seconds (breed, visual ID, triage score)
3. (0:45) Show the database federation query returning "no chip match" — PawLink auto-searches visual DB → shows owner match with 94% confidence
4. (1:10) Owner receives SMS: "Your dog may have been found at Dierenambulance Den Haag. Tap to confirm."
5. (1:30) Show shelter dashboard: live intake queue, capacity forecast chart, adoption matching for recovered animals

Technical Architecture (Scalable)



EC-Relevant Extension (Post-Hackathon)

The Electronic Components angle — cited in your background — naturally extends PawLink into hardware:

- **BLE-beacon collar tags:** Low-energy broadcast of a pet ID code detectable by any PawLink-enabled phone nearby
- **RFID scanner integration:** A unified reader covering 125 kHz, 128 kHz, and ISO 134.2 kHz bridging the legacy frequency fragmentation in existing chips^[42]^[33]
- **IoT kennel sensors:** Temperature, motion, and feeding sensors in shelter kennels feeding into PawLink's care timeline — aligned with livestock IoT research^[43]^[44]

Competitive & Ecosystem Landscape

Platform	Focus	Weakness vs. PawLink
Finding Rover ^[^23]	Consumer facial recognition for lost pets	Consumer-side only; no shelter intake workflow
PetBiometrics.ai ^[^21]	Noseprint ID, cloud inference	Not integrated with shelter operations or notifications
Amivedi ^[^34]	NL lost/found registry	Text-only, passive, no AI, no push alerts
CatRadar ^[^16]	Social media ads for lost cats	Paid per use, not shelter-native, no biometrics
PetAlert.nl ^[^15]	Community alert network	No AI, no shelter integration, manual posting
Booking Ninjas ^[^17]	Shelter management (Salesforce)	No AI identification, no biometrics, costly

PawLink's Unique Position: The only platform designed from the shelter's perspective, combining biometric identification, federated database queries, automated owner alerts, and operational intelligence in one open-source stack.

Eligibility & Scalability Check

Criterion	Status
IMC IM CHANGE eligibility (non-profit beneficiary)	✔ Dierenambulance Den Haag (ANBI-eligible foundation)
Apolitical, non-religious	✔ Animal welfare
Youth/education/inclusion benefit	✔ Supports community volunteers and youth engagement in animal care
Scalable beyond one organization	✔ Open API, multi-org dashboard, national DB federation
Commercializable (secondary)	✔ SaaS licensing for larger shelter chains, insurance integrations, vet platform white-labeling
GDPR compliant	✔ Architecture built on EU data principles, opt-in biometrics

Conclusion

PawLink addresses a real, quantified problem — ~1,000 animals/month in a single branch, 44,000 calls annually — with a technically sophisticated but buildable solution. It sits at the intersection of computer vision, federated data systems, real-time notifications, and IoT hardware — precisely the domain sweet spot of a two-person CS/EC CESE team. It qualifies cleanly for the IMC IM CHANGE challenge, aligns with Dutch and EU regulatory momentum on pet identification, and is architected for scale from day one. The 24-hour MVP is achievable and demo-ready; the long-term platform has a credible path to becoming the infrastructure layer for animal rescue across the Netherlands and beyond.^{[2][3][6][7]}

Sources: HackDelft 2026 presentation, Dierenambulance Den Haag public data, Dutch government pet welfare regulations, [PetBiometrics.ai](#), Finding Rover, Amivedi NL, CatRadar, [PetAlert.nl](#), animal shelter capacity research, AI veterinary triage, pet tech market data, Netherlands veterinary services, Netherlands pet population data, RFID microchip standards.^{[45][39][5][3][33][29][23][4][38][31][34][15][20][6][7][37][1][2][8][21][9][16]}

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